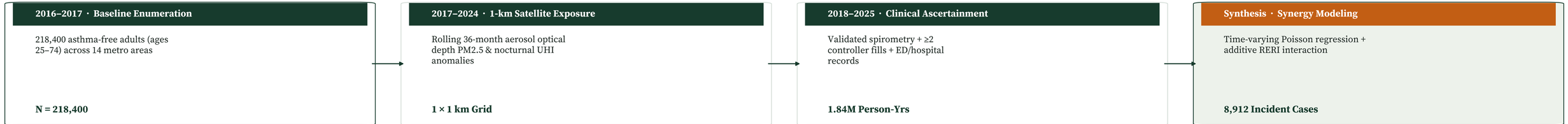
2016–2017 · Baseline Enumeration
218,400 asthma-free adults (ages 25–74) across 14 metro areas
N = 218,400
2017–2024 · 1-km Satellite Exposure
Rolling 36-month aerosol optical depth PM2.5 & nocturnal UHI anomalies
1 × 1 km Grid
2018–2025 · Clinical Ascertainment
Validated spirometry + ≥2 controller fills + ED/hospital records
1.84M Person-Yrs
Synthesis · Synergy Modeling
Time-varying Poisson regression + additive RERI interaction
8,912 Incident Cases